Project Report

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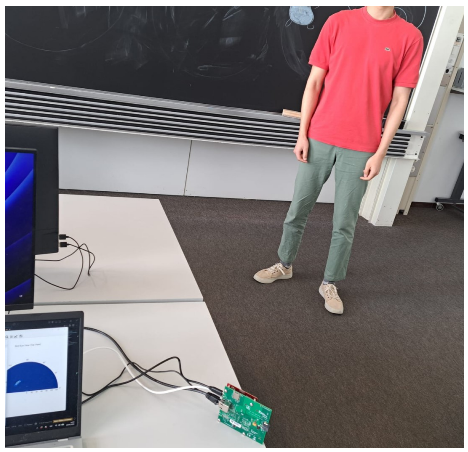


Figure 1: Working with the radar

1 Achivements

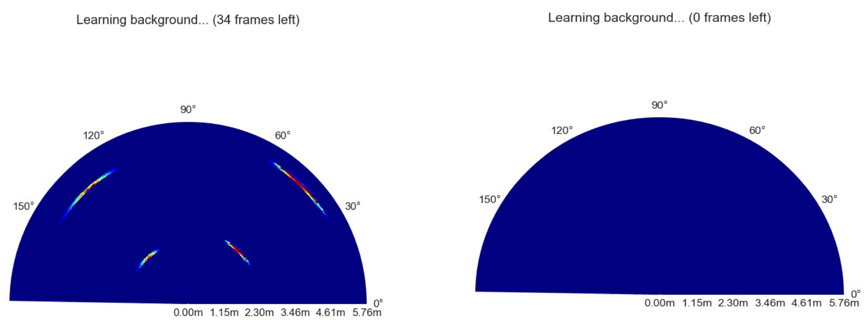
1.1 Refactored architecure

Starting from the initial hands-on implementation, we redesigned the architecture to optimize the pipeline and improve overall robustness. The system is now decoupled into three core components: a producer, a processor, and a visualizer, each running on a dedicated CPU core to minimize latency. Two synchronized queues enable efficient frame transfer between these workers.

1.2 Implemented pipeline

In the processor, we implemented the complete processing pipeline, which performs the following steps: initial environment scanning and background removal, CFAR detection, beamforming, normalization, and finally transmission of the processed frame to the visualizer.

We developed an intelligent background removal method. During initialization, the system scans the environment over the first 50 frames, computes their average, and uses it as a background reference. Each subsequent frame is then subtracted from this reference background. Combined with CFAR detection, this approach performs effectively and allows the system to clearly highlight only the regions where people are present.



(a) Background Scanning

(b) Background Removed

Figure 2: Visual comparison of the processing stages.

1.3 Multitracking

We also attempted to create a 2D point map in addition to the bird's-eye view radar display, where each point on the map represents a detected person. For this purpose, we used DBSCAN for clustering and GTrack for multi-target tracking. However, due to the limited performance of the radar, the obtained results were not sufficiently meaningful or reliable.

2 Status and deviation from proposal

Overall, the project remains aligned with the original proposal and is progressing on schedule. However, the actual development process has differed from the initially expected timeline. Indeed, instead of starting entirely from scratch and implementing each module sequentially, we began from an already functional base (the hands-on project) and integrated multiple components simultaneously.

As a result, the original project timeline is no longer fully representative of our current workflow or progress. Nevertheless, the main objectives remain unchanged, and the project is advancing in the intended direction. However, we recognize the need to accelerate development in order to meet the upcoming milestones.

2.1 Chosen extensions

As mentioned in the project proposal, we indicated that the body recognition extension might be dropped, as it is particularly time-consuming and would require the integration of machine learning techniques. We now confirm that there will not be sufficient time to implement this extension.

We will therefore focus our efforts on the two remaining extensions: through-wall detection and fall recognition.

2.2 Final demonstration objective

The goal, as outlined in the project proposal, is to present a concrete use case of our system to the class during the final week. The demonstration will include examples such as automatically turning lights on and off when someone enters or leaves a room, as well as dynamically adjusting systems based on the number of people present. For instance, this could include controlling air conditioning to reduce energy consumption, improve user comfort, and support greater sustainability.

Additionally, we will present the fall detection functionality using a scenario involving an elderly person falling.

To support the demonstration, we will likely integrate an Arduino Uno connected to the computer to simulate real-world devices. This setup will allow us to emulate actions such as switching lights on and off or adjusting system parameters based on the number of detected people in the environment.

3 Problems encountered & solution

So far, we have focused extensively on optimizing performance before scaling up the system, as it is essential to first ensure that multi-target tracking operates reliably.

3.1 Duplicated target

One issue we encountered is that when a target is detected, we consistently observe a duplicated detection approximately one or two meters behind the actual position. In other words, the system systematically produces a mirrored or secondary point for each real detection on the plot.

At this stage, the cause of this phenomenon is not yet fully understood. We are currently investigating the origin of this duplication in order to identify and eliminate it.

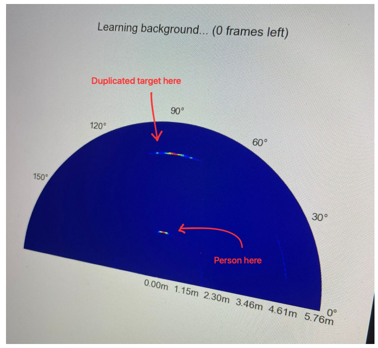


Figure 3: Observation of the duplicated target issue

3.2 Fighting agaisnt noise

Although we achieved meaningful results, we are still working to reduce the noise levels in our plots. Additionally, the radar's detection capability was dropping off significantly beyond 2.5 meters. This was mainly due to two reasons:

- **Signal Scaling and Normalization:** To visualize the data, we normalized our values on a scale from 0 to 1. However, the initial scaling method suppressed weaker signals at further distances. While applying a square root transformation helped amplify these distant detections but also proportionally increased the noise, making it sometimes difficult to distinguish legitimate targets from background interference.
- **Low Radar Cross Section (RCS) of Humans:** Our testing relied on human subjects, who have significantly lower reflectivity compared to metallic objects. Because the human body absorbs and scatters radar energy rather than reflecting it directly, the return signal is roughly 10 to 100 times weaker than metal. This resulted in a much lower signal-to-noise ratio (SNR) as the distance from the sensor increased.

3.3 Configurating 2 radars

We would like to use two radars instead of only one in order to achieve more accurate results for multi-target tracking.

We initially planned to complete this configuration in time for this project report. However, due to complex and time-consuming setup issues on Windows, we were not able to finalize it within the expected timeframe.

4 Next step

There are therefore three major steps remaining for our project: implementing the two-radar system and finalizing multi-target tracking, adding the extensions for through-wall detection and the extension for fall detection.

4.1 Work repartition

Mathys will focus on finalizing and optimizing the multi-target tracking pipeline using two radars. Romain will work on through-wall detection, and Adrian will focus on fall detection.

4.2 Fall detection implementation

For the fall detection extension, we plan to use the 3D position data already provided by the pipeline. We will track each target over time and compute simple motion features such as vertical velocity and acceleration.

A fall will be identified by detecting a sudden and significant downward movement along the Z axis, followed by a period where the person remains stationary close to ground level. This temporal pattern is used as the main indicator of a potential fall event.

This approach is intentionally based on explicit rules in order to ensure real-time performance and ease of integration with the existing tracking pipeline.

4.3 Trough wall detection implementation

The implementation of through-wall detection will begin with a controlled environment using a non-metallic barrier, such as cardboard. We will introduce a hyperparameter to toggle the "wall mode" within our processing pipeline.

To compensate for the signal attenuation caused by the barrier, we must account for the *Two-Way Transmission Coefficient* (L_{wall}). The power received (P_r) from a subject behind a wall is modeled by the modified radar range equation *(refer to the project proposal for a detailed breakdown of all variables)*:

$$P_r = \frac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 R^4 \cdot L_{wall}^2}$$

Our objective is to implement a dynamic gain adjustment in the code that restores the detection sensitivity for subjects behind the wall to levels comparable to open-space environments.

A significant technical challenge involves the **spatial localization of the wall**. The system must accurately identify the wall's distance to avoid over-amplifying reflections from objects or clutter located in front of the barrier. Incorrect amplification could lead to a saturation of the visualizer or a "blindness" to the actual targets of interest. Given that our current detection system remains sensitive to noise, we will focus on refining our CFAR (Constant False Alarm Rate) thresholds specifically for the region beyond the wall's estimated coordinates.